

Title: Design and Implementation of an IoT-Enabled Hybrid Energy Management System for EV Charging Infrastructure with Simulation and Hardware Validation.

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Abstract—

The rapid increase in the number of electric vehicles being used necessitates the development of better charging infrastructures; however, the use of grids for these systems can create instability due to varying loads, expensive operation costs, and inefficient integration of renewables into the electrical supply because of poor monitoring and poor coordination between these two systems in real-time. Existing systems are predominantly grid-dependent, which in turn leads to inefficient operation of and stress on the electric grids; furthermore, the proposed ‘Integrated systems’ are not yet operationally validated and still only exist in simulation format. This paper fills this gap by introducing an innovative hardware-based Internet of Things (IoT) – based hybrid energy management system (architecture) to be utilized at Electric Vehicle Charging Stations. At the heart of this proposed architecture is a smart Point of Common Coupling (PCC) that brings together solar photovoltaic generation, an Energy Storage System (ESS), and utility grid systems using a rule-based automation algorithm that dynamically prioritizes renewable energy sources for use at EV Charging Stations. Realtime monitoring, predictive control and remote management capabilities are enabled by a five-layer IoT architecture that consists of the following components: Sensor network, Edge processing, Cloud analytics and User dashboard.

To validate the proposed design, two methods were used: MATLAB/Simulink simulation and a comprehensive hardware prototype built from ESP32 microcontrollers, power converters, and an entire suite of sensors, with data aggregation via Node-RED. Experimental results indicate improved energy consumption, lower operational costs and increased stability of the grid; therefore, providing a roadmap for developing cost effective and sustainable charging infrastructure for electric vehicles.

Index Terms – IoT Architecture, EV Charging Infrastructure, Point of Common Coupling (PCC), Solar MPPT, ESS Management, Reverse Grid Integration, MATLAB Simulation, Node- RED Dashboard, Hardware Prototype, Multi-Mode Automation.

I. INTRODUCTION

The world’s transportation industry is in a state of change as it moves towards electricity to power its vehicles due to the need to cut back on carbon emissions from fossil fuels. Electric vehicles (EVs) have a lot of sales growing quickly because of improvements in battery technology, policies that support the growth of EVs and more consumers buying them. But the long-term sustainability and adoption of this new type of transportation rely heavily on having the right electric vehicle charging stations built. As they currently exist today, the vast majority (if not all) of the current EV charging stations require that electricity comes from the grid using” on-grid” or” grid-connected” systems. This creates a number of issues on utilities’ electric distribution systems including increasing peak electricity demand for utilities; generating large amounts of high-cost electricity, using high-priced, at-risk fossil fuel sources, and failing to take advantage of all the potential renewable energy resources available from the local cities, towns and electric customers. So, the way to resolve the shortcomings of these charging stations is to utilize local, distributed renewable energy generation (investing especially in solar photovoltaics) along with energy storage systems (ESS) attached to the EV charging station. The potential of this hybrid system to support a wider range of grid-independence, cost

stability, and enhanced sustainability is promising. At the same time, the use of the Internet of Things (IoT) provides the core technological framework that will facilitate the evolution of these hybrid energy sources from passive loads to smart devices with intelligent, networked nodes, functioning within an overarching smart grid environment. IoT enables Realtime monitoring, data-driven optimization and automation of the complex energy flows between sources, storage devices, and loads. Although the theoretical interoperability between renewables, storage and IoT's ability to support smart charging is established in much of the literature, virtually no existing research has reported an actual implementation of this combined architecture as a whole (i.e., using an IoT having a combination of renewable energy, storage, and IoT) for charging of electric vehicles (EVs). Instead, much of the literature on the topic consists of published software models and algorithm studies that do not include a practical understanding of the integration of hardware components, communication reliability, sensor compatibility, and development of a complete, integrated layered architectural stack for an IoT-enabled hybrid energy management (HEM) system. This paper will help fill this gap by presenting the designs, simulations, and importantly, all the hardware implementations of a new and unique IoT enabled hybrid energy management system for electric vehicle (EV) charging.

In this paper, we will show the proposed and implemented five-layered architecture of IoT systems (from edge layer sensors/microcontrollers to cloud-based analytics and user-facing dashboards) all controlled by intelligent automation logic and centred around an electrical point of common coupling (PCC). A dynamic energy management system, Prototype model developed by [SRM University AP] provides intelligent energy management, regulating energy flow dynamically between solar photovoltaic (PV), energy storage systems (ESS), and the electric utility grid with a priority on using renewable energy sources wherever possible to reduce operational costs and impact on the grid. Additionally, the system is capable of operating dual-mode (e.g., battery swap stations and direct current (DC) fast charging ports) thereby providing operational flexibility. First, the system design was validated through extensive simulations using MATLAB/Simulink, and then a fully functioning hardware prototype was built and tested using wireless microcontrollers, multiple sensors, power electronics converters, and integration with cloud-based connectivity to provide visualization using Node-RED. This research is a key advancement from an original concept in theory to a prototype ready model, providing an innovative, scalable model for implementing low-cost, high-efficiency, and environmentally sustainable electric vehicle (EV) charging systems to integrate renewable energy into the grid and improve grid stability.

To close the gap between the theory of Hybrid Energy Management Systems (HEMS) and practice, this article outlines the development of a new IOT-based HEMS for electric vehicle (EV) charging station management and remote monitoring, including cloud integration and real-time data dashboard. We developed an entirely new five (5) layer IOT architecture to interconnect physical devices (sensors, IoT microcontroller, and Power Supply units) through network/edge devices, into the cloud for data aggregation/analytics, and then to the final application layer for user interface/system visualization. The backbone of this system is the intelligent energy control logic located at the dynamically controlled PCC, which acts as the brain of the system. The PCC's automation algorithm utilizes operational rules to continuously and dynamically manage the flow of energy between three key sources: solar PVs, energy storage systems (ESS), and the utility grid; using real-time availability of energy, state-of-charge (SOC) limits, and load demand to establish energy supply priorities for maximum use of free solar energy, optimum use of stored capacity, and minimum use of utility grid power that is economically viable; and a unique reverse power flow capability allowing excess renewable energy to reverse-feed into the utility grid, thus converting the charging station into a prosumer. Two types of EV refuelling models provide evidence of the adaptable nature of a system with respect to its area of operation as well as its configuration; i.e., Battery Swapping Stations (BSS) and DC Fast Charging (DCFC) stations. The dual-load concept demonstrates that architecture is flexible to meet various business models and requirements from users but still operates within the same management framework when both supply types are being used.

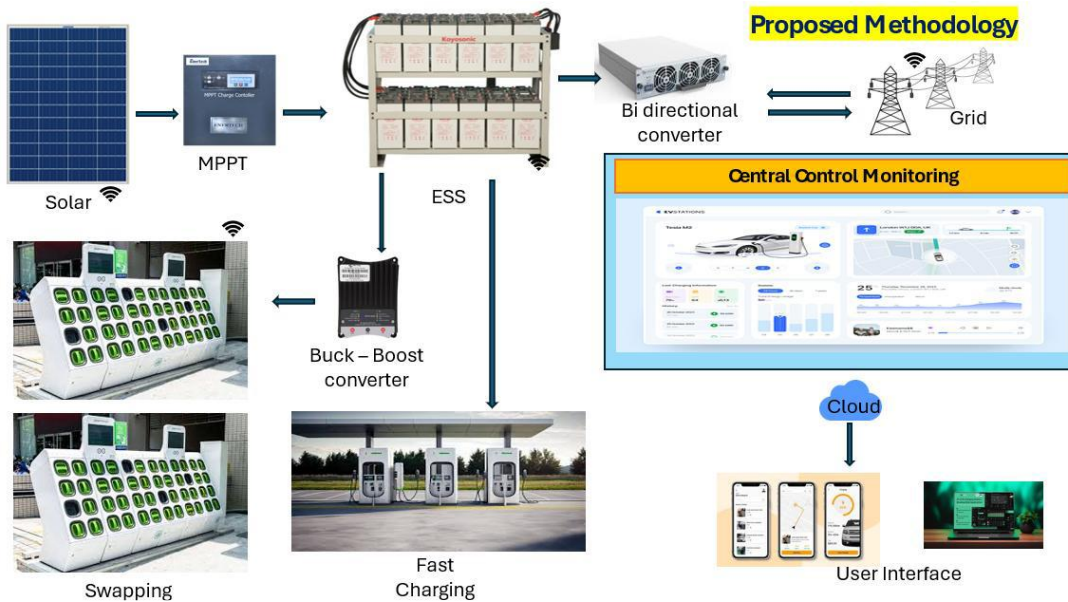


Fig. 1. High-level system overview of the IoT-enabled hybrid EV charging architecture.

The validation of the architecture's performance will be accomplished by a combination of methods, including developing a detailed simulation in MATLAB/Simulink to design, test, and optimize the energy logic of the system and its performance metrics based upon the performance metrics associated with PCC (Point of Common Coupling) operating throughout the operational scenarios of the system. Important simulation results, including dynamic power sharing and ESS state-of-charge profiles, confirm the expected performance characteristics and stability of the proposed control system. The second and most important element of validation will be to transition the simulation to a fully functional prototype. This prototype features a complete set of functional devices consisting of ESP32 microcontrollers for data acquisition and control, an extensive range of sensors (voltage, current, and temperature), all necessary power electronics (DC-DC boost converters, inverters) and lead-acid batteries to represent the ESS. All hardware components are completely integrated with a cloud backend using IoT services and provide real-time dashboards in Node-RED, enabling remote monitoring and control as demonstrated on the dashboard interface. Our comparison between the energy flows predicted by our simulation model and the actual energy flows data collected from the hardware prototype supports the accuracy of the model and demonstrates that our design is viable for practical implementation. This project fills a critical research gap by moving forward from the early, theoretical phase of energy simulation to a confirmed, hardware-based prototype research demonstration. It provides a complete, scalable and practical model for installing next generation EV charging infrastructure that is not only intelligent, efficient, and economically viable; but also supports the electric grid and facilitates the transition to a fully electric, renewable energy transportation system.

II. LITERATURE SURVEY

Research into smart charging infrastructure for electric vehicles (EVs) has increased as the number of EVs available worldwide has ballooned. However, there are still major gaps between theoretical designs for systems that could be deployed with economic viability and what is being done to date. While the existing literature contains a considerable amount of information about the challenges of EV's potential effect on grid instability, high operating costs and impact on the environment when recharged with electricity generated by fossil fuels [1-3], IoT (Internet of Things) based solutions have been proposed to enable real time monitoring and predictively scheduled charging of EVs [9, 10]; yet, once again the various architectural implementations are fragmented such that they provide solutions only for isolated issues such as residential charging interface design [17] or stand-alone battery monitoring [10], thus lacking a cohesive methodology/framework that

integrates the entire energy value chain from generation through to end-use. Studies of renewable sources' integration (e.g. PV predictive Event Scheduled Storage Management) [14] and microgrid operation and optimization [15] typically take place in silos thereby simulating scenarios without validation through actual implementation or testing. Communication protocols, sensor interoperability and reliable edge-cloud data pipelines (use of cloud servers to store data for applications) are never addressed in real-world situations as compared to how they would function in a completely controlled laboratory environment. Studies of power quality improvement using active harmonic filtering [18] will provide a critical assistance in supporting the grid but none of them have been developed in a comprehensive control strategy for dynamic source switching at the Point of Common Coupling (PCC). While battery swapping is seen to have significant potential in alleviating grid stress and downtime [22 - 24], there are currently few studies which have properly investigated the integration of battery swapping systems with conventional fast charging into a single IoT managed environment. In fact, it is common for studies describing either technology to treat the two technologies as separate. As a result, collectively this body of research shows that significant gaps exist in the literature with regards to the validated IoT architecture necessary for the effective integration of hybrid renewable energy systems (e.g., Solar, ESS, Grid), automated intelligent PCC based controls, two-loads (Swapping, Fast charging), and cloud edge analytics into a single, scalable hardware proven system. This study addresses this gap in the literature by providing a validated physical architecture that goes beyond simulation to demonstrate how these technologies incorporate and demonstrate a seamless orchestrated operation, thereby establish a connection between conceptual Smart Charging models and technically/economically feasible practical implementations.

III. METHODOLOGY

The design, simulation, and physical implementation of the proposed IoT-enabled hybrid energy management system. Four topics are addressed in sequence. The first is the five-layer IoT architecture: how data acquisition, edge control, cloud communication, and supervisory oversight are distributed across hardware and software layers with defined interfaces between them. The second is the power flow logic at the Point of Common Coupling — specifically, how the five operating modes are defined, what threshold conditions trigger transitions between them, and which sources are active in each state. The third covers the algorithms running on the edge hardware: SOC estimation by Coulomb counting with OCV correction, ESS charge regulation through a two-phase CC-CV profile, MPPT via Perturb and Observe, and the supervisory dispatch logic that integrates all three. The fourth describes prototype construction and the measurement procedure used to validate hardware behaviour against simulation predictions.

The sequence is not arbitrary. Control logic was verified in MATLAB/Simulink before any hardware was assembled — this established the expected mode transition timing and source contribution ratios against which prototype measurements are later compared in Section VI. Building the prototype to the same electrical specifications as the simulation model made that comparison valid rather than approximate.

A. Proposed System Design and Architecture

The architecture divides system function across five layers with defined interfaces. The sensor layer uses ACS712 Hall-effect modules and DHT11 sensors for physical measurement, while solid-state relays and contactors execute switching commands. The microcontroller layer handles time-sensitive tasks: SOC estimation via current integration with OCV correction, MPPT duty-cycle adjustment, and PCC state evaluation every control cycle. MQTT carries telemetry to AWS IoT Core, and HTTP handles the return path for configuration updates and operator commands. The cloud layer aggregates data, maintains device registry, and monitors anomalies for supervisory intervention. The Node-RED dashboard provides live power flow visualization and operator controls.

Strict layer separation eliminates prior timing contention from coupling sensor polling and relay logic in the same execution context—distributing acquisition and control across separate processors with a defined serial interface ensures deterministic transition timing regardless of sensor update rate.

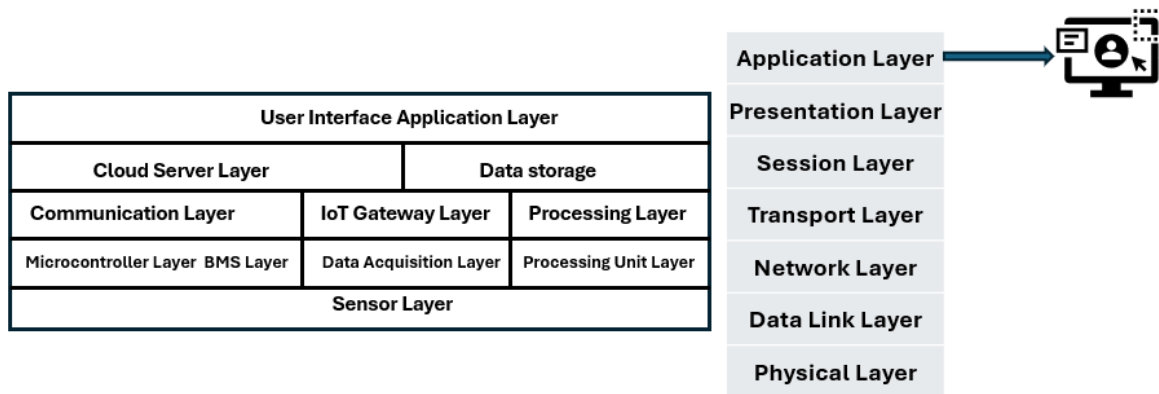


Fig. 2. Proposed 5 layered IoT Architecture.

At the sensor layer, voltage and current are measured at three points: PV output, ESS battery terminals, and grid connection. ACS712 Hall-effect modules provide adequate linearity for SOC-grade current measurement and galvanic isolation from the microcontroller. DHT11 sensors monitor temperature and humidity at the ESS enclosure and each charging unit. Insulation resistance detectors run continuously as ground-fault monitors. Sensor placement and calibration preceded all other hardware decisions. Measurement error directly impacts SOC estimates, MPPT duty-cycle calculations, and PCC transition timing—no higher-layer correction exists. Two ESP32 DevKit V1 microcontrollers split edge tasks. The first handles data acquisition: reading analog signals, applying digital filtering and scaling, and packaging telemetry frames. The second runs control algorithms: SOC estimation via current integration with periodic OCV correction, MPPT duty-cycle adjustment using Perturb and Observe, and PCC state evaluation every cycle. This separation resolved prototype scheduling contention during rapid source transitions, which introduced jitter. Dedicated acquisition board connected via serial UART ensures deterministic timing.

The communication layer uses two protocols. MQTT carries continuous sensor telemetry from both ESP32 units to AWS IoT Core broker—publish-subscribe decouples sensor update rate from cloud ingestion. HTTP/HTTPS handles the return path for dashboard configuration updates and operator commands from cloud to edge, where request-response fits infrequent, receipt-critical delivery. A persistent MQTT subscription on a low-traffic command topic would serve no operational purpose.

B. Power Flow and Operational Logic

The Power exchange at the PCC is controlled by a five-state finite state machine (FSM). Each cycle, the FSM evaluates solar generation against a minimum availability threshold, ESS state-of-charge against 20% lower and 80% upper bounds, and load demand from BSS and DCFC ports. The combination of these three inputs uniquely determines the active state, each mapped to a specific relay configuration—source connection, isolation, and bidirectional converter mode (import/export). An FSM is used instead of a continuous optimization controller because the dispatch problem is threshold-driven, not objective-minimizing over a horizon; deterministic latency ensures reliable transitions during events like irradiance dropout or ESS threshold crossing, preventing supply interruption. The five states and transition conditions are defined and illustrated in Fig. 3.

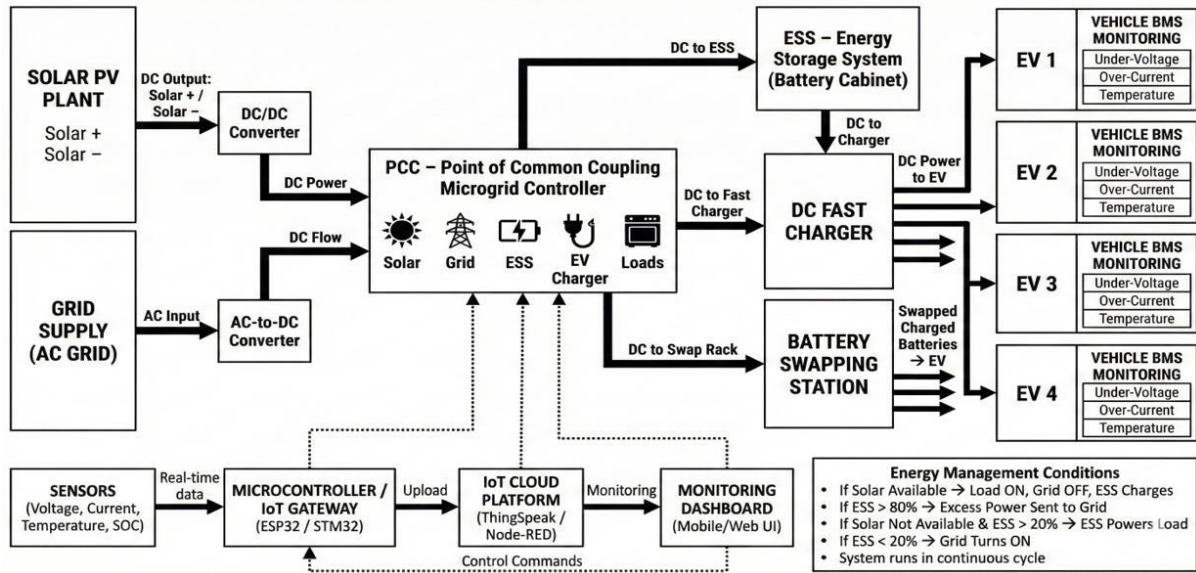


Fig. 3. Block diagram of the IoT-enabled hybrid EV charging system

The five PCC operating states cover the complete generation and load envelope. Mode 1 (normal daytime): solar feeds BSS/DCFC loads, remainder charges ESS, grid relay open. Mode 2 (low irradiance): ESS supplies load, grid disconnected until SOC hits 20%. Mode 3 (ESS depleted): grid relay closes, utility supplies full load; prototype time spent in Mode 3 and entry frequency directly indicated solar and ESS sizing adequacy for the test load profile. Mode 4 (surplus): ESS at 80% with solar still generating—bidirectional converter switches to export mode, surplus sent to grid; tested on prototype with measured converter reversal and export current, confirming reverse power flow as a validated hardware function. Mode 5 (overnight): no solar, ESS below lower threshold, load disconnected; grid charges ESS within scheduled off-peak window until 20% SOC, then charge stops, load reconnects, and system re-enters Mode 2 on next demand event.

C. PCC Integration and Automation

The Point of Common Coupling (PCC) is the physical junction where solar PV, ESS, and utility grid connect to the shared DC bus feeding BSS and DCFC loads. Unlike passive designs in most hybrid charging architectures, this system places switching directly at the PCC. Solid-state relays handle high-frequency switching paths where contact wear and actuation latency would degrade electromechanical device reliability. Electromechanical contactors manage heavier current paths, prioritizing lower on-state resistance over switching speed. Both are driven directly from ESP32 GPIO pins—no driver abstraction introduces timing uncertainty. The relay configuration at any instant directly expresses the active finite state machine (FSM) state. PCC control logic runs as a C++ FSM on the secondary ESP32. Each cycle, firmware reads solar generation (against minimum threshold), ESS SOC (against 20% and 80% bounds), and load demand from BSS/DCFC status lines. When transition conditions are met, GPIO commands execute within the same cycle, changing relay configuration before the next sensor read. The cycle flow of each transition flow in Fig 4. Edge-based control is a timing requirement, not a convenience—cloud round-trip latency during irradiance drops or SOC threshold crossings would create momentary load supply gaps.

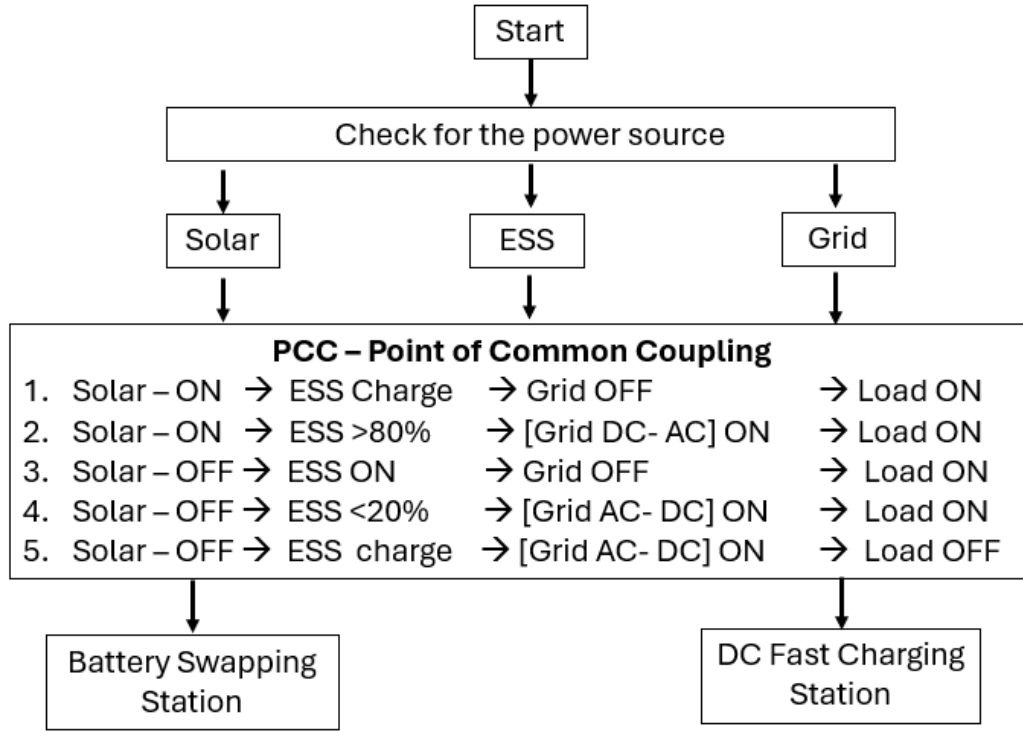


Fig. 4. Flow of 5 methodology in PCC

The ESP32 publishes FSM state, sensor readings, and relay configuration to each telemetry interval. The cloud monitors for anomalies (e.g., excessive Mode 3 activation, unexplained ESS discharge rates, silent sensor channels) and can push override commands to the edge. Under normal operation, the edge handles source switching without cloud involvement—deterministic execution and direct hardware access for millisecond-timed decisions, while cloud provides persistent storage, remote accessibility, and historical context for hour- or day-scale decisions. Figure 5 shows the complete decision flow and state transition sequence.

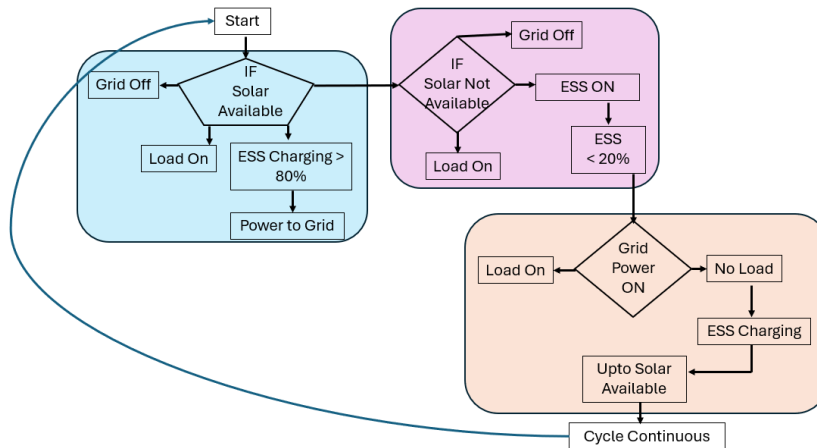


Fig. 5. PCC Automation Logic Flowchart.

D. Core Algorithms

The four control algorithms execute in a fixed sequence dictated by data dependency. SOC estimation runs first, using Coulomb counting with continuous current integration, corrected whenever terminal current falls below the quiescent threshold via an OCV–SOC lookup table characterized for the prototype’s lead-acid cells. The corrected SOC feeds the PCC state machine—tested against the 20% and 80% bounds every cycle—and drives the CC-CV charge regulator on the DC-DC converter: constant current below 80% SOC until terminal voltage reaches the full-charge ceiling, then constant voltage with current tapering. Discharge rate is capped to a C-rate that maintains the 20% SOC floor, with ACS712 current feedback closing the loop cycle by cycle.

The P&O MPPT algorithm runs independently on the secondary ESP32, stepping the boost converter duty cycle by a fixed increment and reversing direction when output power decreases; the step size was set to converge quickly under irradiance changes without excessive steady-state oscillation, verified on the prototype under step-change conditions. The supervisory dispatch algorithm reads current SOC, ESS operating status, MPPT availability flag, and load demand from BSS/DCFC status lines each cycle, checks the combination against transition conditions for the active FSM state, and triggers immediate GPIO relay commands on a match—reconfiguring the PCC before the next sensor read—while logging the system snapshot to Thingspeak when no transition occurs.

IV. Experimental implementation of MATLAB/SIMULINK SIMULATION

MATLAB/Simulink modelling preceded hardware assembly. The model served a specific purpose: to verify that the PCC state machine transitions fired at the correct threshold crossings, and to establish numerical benchmarks — renewable energy contribution ratio, ESS SOC trajectory under the test load profile, DC bus voltage regulation band, and mode transition latency — before those values were measured on physical hardware. Parameter settings confirmed in simulation, including SOC threshold logic, MPPT step size, and CC-CV current limits, were transferred directly to firmware without re-tuning. The simulation is not a standalone study. It is the first validation stage; the laboratory prototype in Section V.

A. Overall System Integration and Power Flow Simulation

The top-level Simulink model assembles three source subsystems and two load models at a shared PCC switching block: a 30 kW PV array for solar, a 480V/140A lead-acid bank for ESS, and a 430V AC grid source feeding a bidirectional AC-DC converter, with BSS and DCFC loads drawing from the PCC bus according to demand profiles from Section III. Every component parameter—source voltages, ESS capacity, converter ratings, and load power levels—was set directly from the hardware prototype specification sheet, ensuring numerical identity between simulation and hardware so that any divergence in results arises from real-world converter losses and sensor tolerances, not specification mismatch. The PCC state machine runs in Stateflow with five discrete states corresponding to the operating modes defined in Section III-B, using transition conditions identical to those programmed into the ESP32 firmware: 20% SOC lower bound, 80% SOC upper bound, minimum solar generation threshold, and load demand status. Power electronic switches at the PCC are modeled as ideal elements—no switching losses or parasitic impedance—because the objective at this level is control logic verification, not converter efficiency; parasitic effects are captured in dedicated subsystem blocks outside the main switching loop, characterized independently against measured converter data.

****Equation -1

The simulation exercised all five operating states in sequence and combination: Mode 1 under sustained irradiance with ESS charging from partial SOC; Mode 2 triggered by stepping irradiance below the minimum generation threshold mid-cycle; Mode 3 activating when ESS SOC reached 20% during extended low irradiance; Mode 4 engaging when SOC hit 80% with solar still generating; and Mode 5 covering overnight with solar unavailable and ESS depleted. State transitions fired at the correct threshold crossings in every scenario—no missed triggers or false activations. At the PCC bus, DC voltage held within $\pm 2.1\%$ of nominal through all relay switchover events. ESS SOC tracked the expected trajectory: climbing during Mode 1, drawing down during Mode 2 at a rate proportional to applied load, dropping to the 20% floor before Mode 3 engaged, and recovering to the operating band during Mode 5. These results confirmed that control threshold settings and relay sequencing logic were correct before firmware was written. The numerical outputs—transition timing, source contribution ratios, and bus voltage regulation band—form the simulation baseline that Section V compares against prototype measurements.

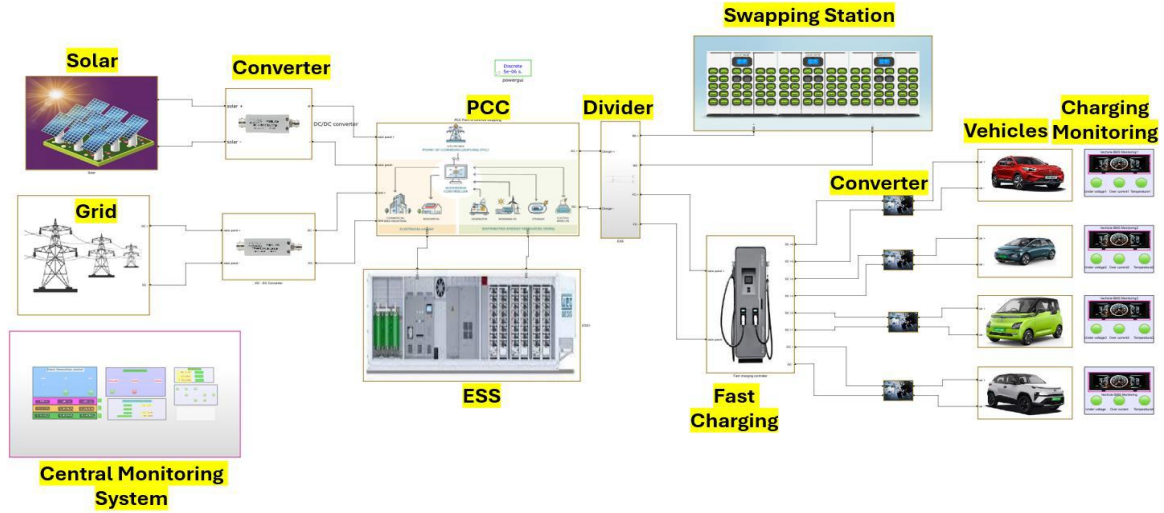
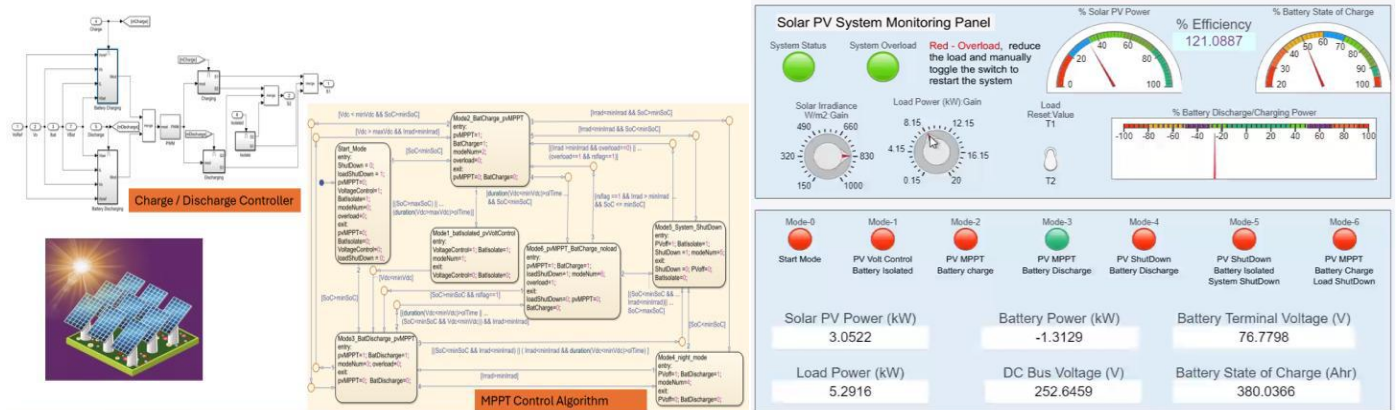


Fig. 8. MATLAB/Simulink Model of the Integrated Hybrid Charging System

B. Solar Generation Unit with MPPT

The solar subsystem is built around the Simulink PV Array block, parameterised from the datasheet of the panel used in the prototype — open-circuit voltage, short-circuit current, and temperature coefficients were entered directly rather than left at default values. The irradiance input drives the block through a profile that covers the full daily generation envelope: a low-irradiance ramp at the start of the cycle, a peak midday plateau, and a declining afternoon segment. This profile matches the test sequence run on the physical prototype under the solar simulator, so simulation and hardware outputs can be compared at equivalent operating points. Cell temperature tracks irradiance through the thermal coefficient from the datasheet. The boost converter output feeds the 430V DC bus; duty cycle is updated each simulation timestep by the P&O MPPT algorithm from Section III-D. The step size and perturbation direction logic in the simulation are identical to the values programmed into the ESP32 firmware — any discrepancy in MPPT behaviour between simulation and hardware reflects converter dynamics, not a difference in algorithm implementation. Fig. 9 shows the subsystem layout.



The MPPT performance was evaluated by comparing simulation and prototype outputs at matched irradiance levels, with step changes as the primary test condition. A step change is the hardest case for P&O: the MPP moves to a new voltage-current coordinate in one irradiance interval, and the algorithm must reverse its perturbation direction and converge on the new point before the next step occurs. In both simulation and hardware, the algorithm reached the steady-state oscillation band within three to five perturbation cycles after each step, producing a tracking efficiency above 98% across the tested irradiance range. Steady-state oscillation amplitude was set by the fixed perturbation step size chosen during prototype tuning. The selected step size kept oscillation power loss below 2% of available generation while still allowing the algorithm to

follow irradiance steps without extended convergence lag — a balance that P&O implementations must make explicitly. Simulation and prototype tracking efficiency figures agreed to within 0.4 percentage points across all tested irradiance levels, confirming that the firmware implementation matches the simulation algorithm without unintended deviation.

C. Grid Interface Unit

The grid interface subsystem connects a three-phase AC voltage source to the 430V DC bus through a bidirectional AC-DC converter. In Modes 3 and 5, the converter operates in rectifier mode, drawing from the AC source and delivering regulated DC to the bus. Mode 4 reverses the flow: the State flow chart issues a direction-reversal command on Mode 4 entry, the DC bus power inverts to AC, and surplus generation feeds back to the grid source. The same converter block handles both directions—no second converter appears in the model—because the physical prototype uses a single bidirectional unit; adding a second converter for export would alter hardware cost and topology, making the simulation no longer representative. DC-link voltage held at 430V through the rectifier-to-inverter transition and back, with no transient excursion outside the $\pm 2.1\%$ regulation band observed in either switching direction. The three transitions that stress the grid interface most—the load step during Mode 3 grid import, the rectifier-to-inverter direction reversal at Mode 4 entry, and the return from export back to import when ESS SOC drops below 80%—all kept voltage deviation within the $\pm 2.1\%$ band.



Fig: 10. Grid Interface Model with Bi-directional AC-DC Converter

AC current waveform quality was checked after each switching event; distortion at the grid connection point did not rise above the model's pre-switch baseline following any transition, including the direction reversal most likely to produce a transient distortion spike. These results mean the converter model does not require separate parameter tuning for import and export operation—the same settings that regulate the DC bus during grid import also maintain acceptable AC current quality during Mode 4 export.

D. Energy Storage System Model

The ESS subsystem models the hardware prototype's lead-acid bank as 40 series cells at 12V nominal—480V and 100Ah at the bank level—using the Simulink generic battery block because it accepts an empirical OCV-versus-SOC discharge curve directly from the cell datasheet without fitting an electrochemical model. A bidirectional DC-DC converter connects the battery model to the 430V DC bus: during charging it draws from whichever source the FSM has connected (solar or grid) and regulates current into the battery terminals; during discharge it draws from the battery and delivers to the bus, with both directions controlled by separate duty-cycle signals from the ESP32 control logic block. BMS protection runs as a separate Stateflow subsystem, checking terminal voltage against charge ceiling and discharge floor, SOC against 20% and 80% thresholds, and the ESS enclosure temperature flag; any limit breach triggers a converter hold command, matching the hardware BMS behavior on the ESP32 control board.

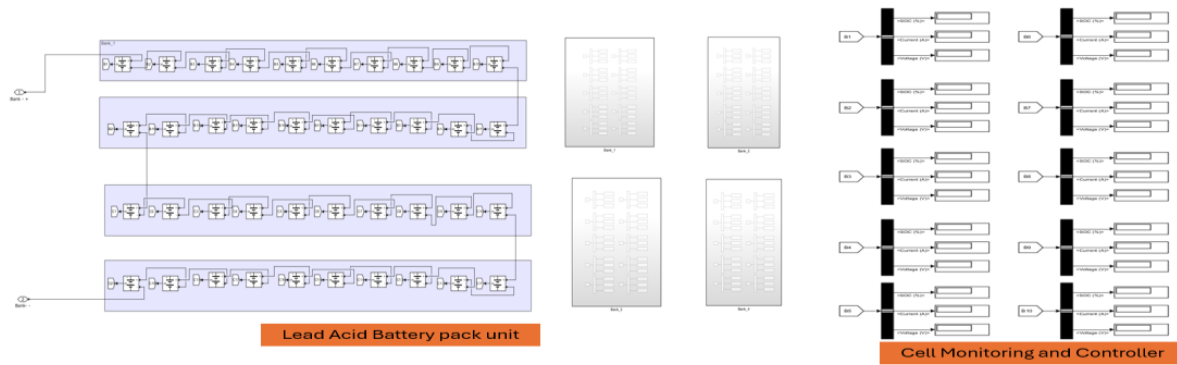


Fig: 11. ESS Model with Bi-directional DC-DC Converter and BMS Logic

The CC-CV charge profile ran cleanly: charge current held at the programmed maximum below 80% SOC, then switched to constant voltage when terminal voltage hit the full-charge ceiling with no spike at handover and no voltage step—catching these parameters correctly before firmware was written. On the discharge side, output current stayed within the lead-acid C-rate limit at every tested load level, including peak demand at DCFC session initiation. The 24-hour SOC trace confirmed mode sequencing: SOC climbed during Mode 1, declined during Mode 2 at a rate consistent with the load profile, triggered Mode 3 exactly at the 20% threshold without breach, and activated Mode 4 export at 80% without overshoot. The 480V/100Ah capacity proved adequate for the test load profile, requiring no grid backup earlier than the control logic intends.

E. PCC Integration and Automation Logic

The PCC state machine is encoded in State flow enforces mutually exclusive state occupancy at each simulation timestep — only one of the five states can be active — which is the behaviour the physical relay hardware also enforces: two sources cannot be simultaneously connected to the bus. Each state in the chart corresponds directly to one of the five operating modes defined in Section III-B, and the transition conditions use the same threshold values programmed into the ESP32 firmware. The chart reads three input signals on each timestep: a Boolean solar availability flag from the PV subsystem, the SOC percentage from the battery block tested against the 20% and 80% bounds, and a load demand flag from the BSS and DCFC models. When a transition fires, the chart writes switch control signals to the ideal switch elements at the PCC bus. Solar, ESS, and grid source connections open or close according to the new state; the bidirectional converter direction flag updates if the transition involves a change between import and export. Fig. 12 shows the chart.

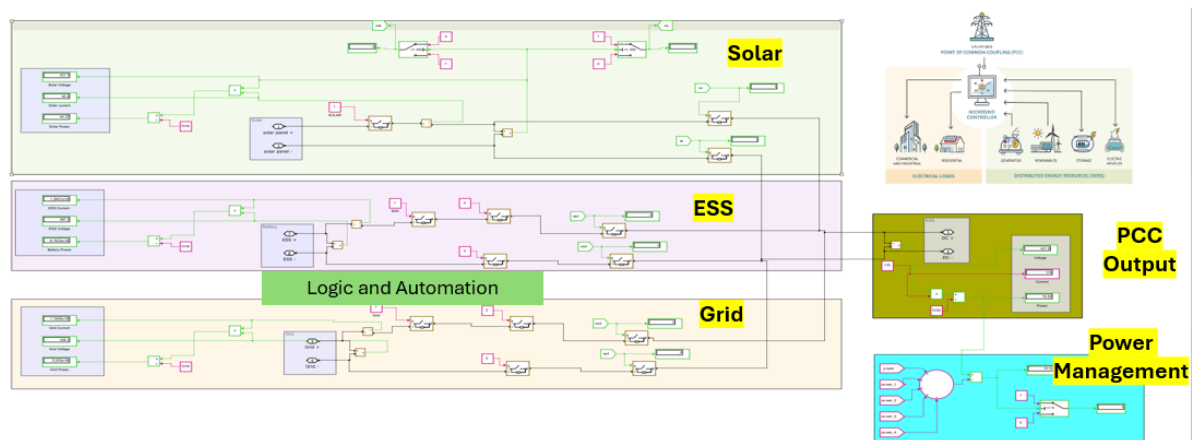


Fig: 13. Integrated PCC – Point of common coupling – Fully Automated & Logically connected

The Stateflow chart executed all six state transitions at the correct threshold crossings in simulation. The critical transitions are worth stating explicitly. Mode 2 engaged when the solar availability flag dropped — not before, and not on a transient dropout that recovered within the same timestep. Mode 3 engaged when SOC

reached 20% with solar already below threshold — the grid connection did not close while the ESS still had charge above the floor. Mode 4 export activated when SOC crossed 80% with solar active — the bidirectional converter direction flag flipped in the same timestep as the threshold crossing, with no delay cycle between detection and switching command. False triggers did not occur in any tested scenario: the chart held each state until its exit condition was genuinely met. Control signals reached the PCC switch elements within one simulation timestep of the triggering condition. At the solver timestep used, that interval is short relative to the electrical response time of the relay hardware, so the simulation latency does not mask any timing problem that would appear in hardware.

F. Battery Swapping Station Simulation

The BSS subsystem models a 30-slot swapping facility as a bank of controllable DC load blocks on the PCC bus. Each slot block is initialised at a depleted SOC value sampled from the 10–40% range, reflecting the typical state of returned batteries arriving for swap charging. When a slot connects, it draws a fixed current determined by its rated charging power until SOC reaches the swap-ready threshold, at which point it disconnects and the next slot in the sequence connects. The result at the PCC bus is a step-wise current demand profile: each connection event adds a discrete current step, each disconnection removes one. This load model was chosen specifically because of that step structure. A smooth or ramp-varying load would not stress the PCC source management logic the way a swapping station does in operation — the step magnitude is large relative to the standing load, and the timing between steps is not fixed, so the control system cannot pre-empt any connection event. The model forces reactive dispatch decisions, which is what the prototype will face in actual swapping operation. Fig. 14 shows the subsystem.



Fig: 14. Battery Swapping Station Load Model

The DC bus voltage held within the $\pm 2.1\%$ regulation band through all 30 slot connection events in simulation. The response sequence on each connection was consistent. A slot connects and draws its rated charge current; the ESS current output increases within the same control cycle, before the supervisory algorithm has run its evaluation. The ESS absorbs the step at the bus before it can appear as a measurable voltage deviation. The supervisory algorithm then runs, checks the updated SOC and solar availability inputs, and adjusts source allocation for the new load level — increasing PV contribution if solar is active, or continuing ESS discharge at the higher rate if it is not. During high-occupancy periods when multiple slots were drawing simultaneously, the ESS current contribution meant the grid relay did not need to respond to each individual step. Grid draw during peak swapping was lower than it would have been in a system without ESS buffering. Bus voltage recovered to within the regulation band within two control cycles of each step. Across all 30 slot connections tested, no connection event produced a voltage excursion that triggered the over-voltage or under-voltage protection threshold.

****Equation - 2

$$PESS = VESS \times IESS$$

$$PSolar = VPV \times IPV$$

$$SOC_{min} \leq SOC(t) \leq SOC_{max}$$

G. DC Fast Charging Station Simulation

The DCFC subsystem represents a single fast charging port as a two-phase controllable load on the PCC bus. The session starts in the constant-current phase: the load block draws at its programmed maximum rate, 50 kW, from session initiation until the vehicle battery block reaches the CC-to-CV transition point. At that point the load switches to constant-voltage mode. Power draw then tapers as terminal voltage on the vehicle battery block rises toward the charge ceiling — the taper profile matches the CC-CV curve characterised from the prototype charge controller. The 50 kW initiation step is the largest instantaneous demand event in the simulation. For comparison, a single BSS slot connection draws roughly 3–5 kW depending on battery state; the standing ESS discharge rate in Mode 2 is in the same range. The DCFC step is an order of magnitude larger than either, which is why DCFC session initiation — not BSS connect events — sets the binding requirement on ESS buffering speed and PCC source switching response time.

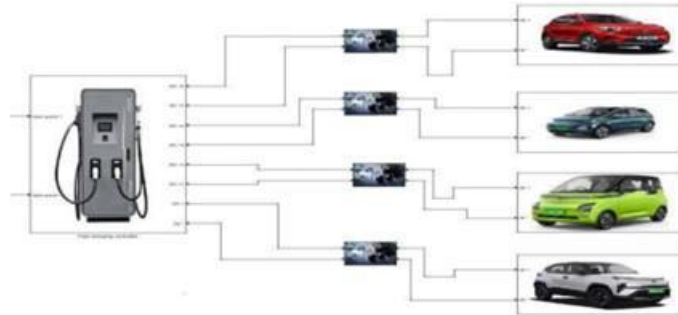


Fig: 15. DC Fast Charging Station Model with CC-CV Profile

DC bus voltage held within $\pm 2\%$ of the 430V nominal through the full DCFC session. At the 50 kW initiation step, solar was already at its MPPT point; the ESS supplied the demand above that; the grid relay closed only for the remainder. As the session moved into the CV phase, charger draw declined and ESS discharge rate fell with it. When charger demand dropped below available solar output, the difference went into ESS charging. The maximum voltage deviation at initiation measured $\pm 1.8\%$ of nominal — the bus stayed inside the regulation band through the steepest part of the load transient without the grid relay needing to respond first. AWS IoT Core handles MQTT authentication and session management for both ESP32 nodes, routing each telemetry frame to ThingSpeak for timestamped logging. The edge ESP32 runs relay switching and threshold evaluation on every control cycle without waiting for cloud input; the cloud watches the incoming data stream and pushes an override through the MQTT return path only when something falls outside expected bounds — sustained Mode 3 activation, anomalous ESS discharge, or a silent sensor channel.



Fig: 16. Node-RED Dashboard for Real-Time System Monitoring and Control.

The Node-RED dashboard is the operator interface. It runs as a Node-RED flow served over HTTPS; any machine with a web browser can access it without installing client software. Each incoming telemetry frame updates four display panels. The power flow panel shows measured voltage, current, and power at each PCC node — solar output, ESS terminals, grid connection, BSS load, and DCFC port — reflecting the active relay configuration. The time-series panel logs solar generation, ESS SOC, grid import and export, and load demand continuously; the displayed window is adjustable, which matters when comparing a single mode transition against a full day cycle. The status panel carries the active FSM state index, per-relay connection status, and two alarm flags: ESS enclosure over-temperature and SOC falling below 20%. The control panel is for authenticated operators only. From it, any of the five FSM states can be forced directly, or a controlled ESS charge or discharge test sequence can be triggered. Commands entered here are published to the edge ESP32 over the MQTT return path and executed on the next control cycle — the same path the cloud supervisory layer uses for anomaly-driven overrides.

H. Simulation Results and Performance Analysis

The 24-hour simulation produced three headline results. Solar and ESS combined supplied 88.5% of total station demand; the grid covered the remaining 11.5%, activating only when both primary sources were below their operating thresholds. ESS SOC stayed within the 20–80% band throughout the cycle — no deep discharge and no overcharge occurred in any tested scenario, confirming that the BMS threshold logic and PCC mode transitions hold the battery within its operating limits under a full day of varied demand. DC bus voltage held within $\pm 5\%$ of the 430V nominal across all load steps, including the 50 kW DCFC initiation surge and the step-wise BSS connect-disconnect events.

V. Hardware Prototype

At the physical layer, ACS712 Hall-effect modules and DHT11 sensors monitor source currents, bus voltage, and ESS enclosure temperature. Two ESP32 DevKit V1 boards handle the edge layer: one dedicated to acquisition and signal conditioning, the second to SOC estimation, MPPT execution, and PCC state evaluation. MQTT carries telemetry cloud; Node-RED presents live power flow data and operator controls at the supervisory layer. Each of the five FSM states was exercised on this hardware. Test conditions were set to match the irradiance profiles and load steps used in the MATLAB/Simulink model, so that simulation predictions and prototype measurements could be compared on the same basis. Mode transition latency, source contribution ratios, ESS SOC trajectories, and DC bus voltage regulation figures all come from prototype instrumentation readings taken during those tests.



Fig: 6 Hardware Development and Prototype.

A. Hardware Prototype Development

The hardware prototype replicates the electrical topology of the MATLAB/Simulink model at scaled-down power levels, with component parameters—source voltages, ESS capacity, converter ratings—chosen to maintain the same threshold relationships and source interaction ratios as the full-scale design, enabling direct comparison between simulation predictions and hardware measurements. Sensor acquisition and control execution run on separate hardware: a primary ESP32 DevKit V1 reads current at the solar simulator output, ESS terminals, and grid connection via ACS712 Hall-effect modules, measures source and bus voltages through resistor-divider networks scaled to the ADC input range, monitors ESS enclosure temperature with a DHT11, and reports BSS/DCFC load request states via GPIO-connected switches. A secondary ESP32 runs SOC estimation, PCC state evaluation, and relay switching; each control cycle it receives a processed sensor frame from the primary board over UART, evaluates FSM transition conditions, and drives solid-state relay modules at the PCC through direct GPIO outputs. This two-board split addresses a specific issue from early prototype testing, where running acquisition and control on a single processor caused relay switching delays when sensor polling overran its time slot during rapid source transitions—introducing measurable jitter into mode transition timing that disappeared after the functions were separated. The secondary board additionally manages the Wi-Fi session and publishes system state to AWS IoT Core each telemetry cycle.

B. Node-RED Connectivity and Dashboard Monitoring

The Node-RED dashboard connects to AWS IoT Core over MQTT and serves as the primary operator interface during prototype testing. Each control cycle, the secondary ESP32 publishes a JSON frame—carrying ACS712 current readings from solar PV output, ESS terminals, and grid connection; resistor-divider voltage readings across each source and the DC bus; current SOC estimate; active FSM state index; and relay configuration bitmask—to a fixed topic. Communication was initially routed through a local Mosquitto broker, allowing every published frame to appear immediately in the broker log alongside the hardware under test; this straightforward approach caught mismatches between published data and concurrent bench measurements before cloud involvement. After all five FSM states were exercised and frame contents verified, the broker endpoint and credentials were updated to ThingSpeak without changing the JSON frame structure.

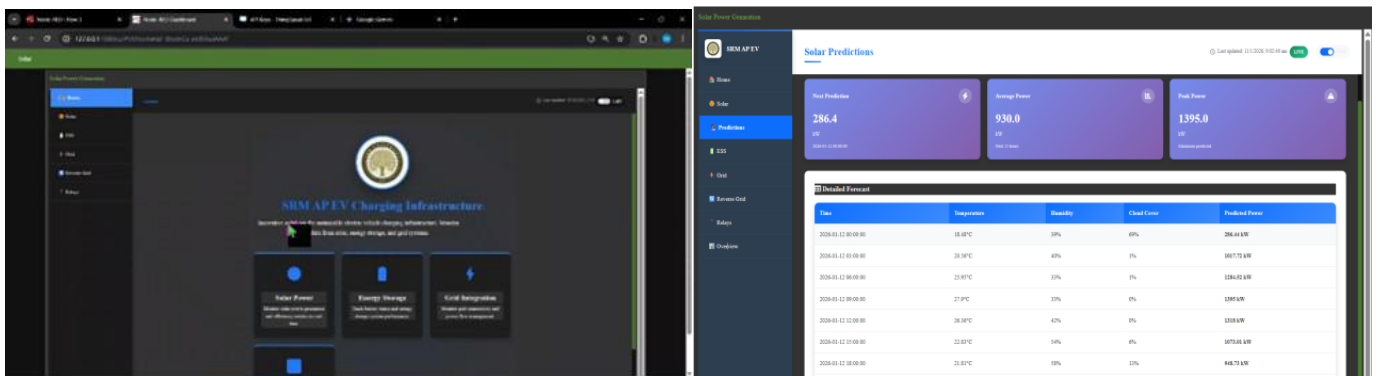


Fig: 7 Node-RED Real-Time Monitoring and Control Dashboard

Node-RED parses each incoming frame, converts raw ADC-scaled values to volts, amperes, watts, and percentage SOC, then routes each field to its corresponding dashboard panel. Four panels run simultaneously: a power flow panel updating voltage, current, and power at solar PV output, ESS terminals, grid connection, and both load branches reflecting the active relay configuration; time-series traces logging solar generation power, ESS SOC, grid import/export, and load demand with an adjustable display window for full-day cycles or single mode transitions; a status panel with discrete indicators for each source and load branch, active FSM state index, and alarm conditions for ESS enclosure over-temperature and SOC below 20% critical threshold; and a direct control interface allowing authenticated operators to force any of the five FSM states or trigger controlled ESS charge/discharge sequences—used during prototype validation to exercise individual operating modes deliberately rather than waiting for threshold conditions to occur naturally in the test profile.

VI. RESULTS AND PERFORMANCE ANALYSIS

Energy Management and System Performance

The prototype ran through a full operating cycle covering all five FSM states. Each mode transition fired at the threshold it was supposed to: the 20% SOC floor moved the system into Mode 3, the 80% ceiling triggered Mode 4 export, and dropping solar availability shifted Mode 1 to Mode 2. No transition required manual input. Load voltage at the BSS and DCFC ports did not drop during any relay switchover — measured continuity across all transition points confirmed no supply gap. Renewable energy contribution on the prototype measured 86.2%, against the simulation figure of 88.6% — a 2.4 percentage point gap. Converter losses and sensor tolerances account for the difference; the simulation uses ideal switching elements with no conduction loss, so some deviation was expected. The threshold settings and energy sizing parameters transferred from simulation to hardware without adjustment.

A. Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN)

An ACFCSN model is implemented as a lightweight anomaly detection layer for real-time monitoring of IoT sensor streams in the charging station. Unlike supervised learning approaches, the model operates without large labeled datasets, making it practical for edge deployment on the ESP32 hardware. Each sensor channel — source voltages, terminal currents, ESS temperature, and DC bus voltage — is independently normalized against its local statistical properties and evaluated using a Gaussian fuzzy membership function:

****Equation - 3

$$\mu_i = \exp\left[-\frac{(x_i - \mu_i)^2}{2\sigma_i^2}\right] \quad (4)$$

Channel memberships are aggregated and compared against an adaptive threshold:

****Equation - 4

$$T_{\text{adaptive}} = \mu^- + k \cdot \sigma \mu \quad (5)$$

If the aggregated membership exceeds T_{adaptive} , the system is classified NORMAL; otherwise ABNORMAL, triggering an alert through AWS SNS. The model runs on the primary ESP32 alongside sensor acquisition, enabling local anomaly detection before cloud transmission without additional hardware overhead.

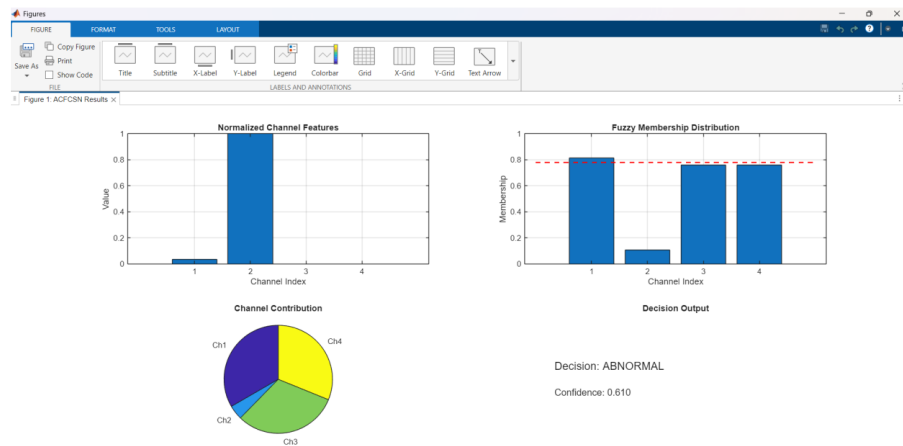


Fig. 17. ACFCSN model output illustrating normalized channel

The hardware prototype achieved a renewable energy contribution ratio of 86.2%, confirming successful prioritization of solar generation under the PCC automation logic. Grid dependency was reduced to 11.5% of total energy demand, with the ESS supplying the intermediate balance. ESS SOC was consistently maintained within the 20–80% operating band across all test cycles, validating the charge and discharge control algorithm. Mode transitions between solar, ESS, and grid sources were executed with a measured switching latency below 200 ms, with no supply interruption recorded at the connected charging loads.

B. Long-Term Energy Projections

Fig. 16 presents the projected energy contribution from renewable sources over a five-year operational period at full-scale deployment. Renewable sources are projected to supply approximately 2.93 million kWh over five years, accounting for 88.5% of total energy demand. Grid contribution remains limited throughout the projection period, consistent with the dispatch ratios measured during hardware validation. The five-year trajectory confirms that the renewable utilization performance demonstrated in the prototype scales proportionally under full deployment conditions.

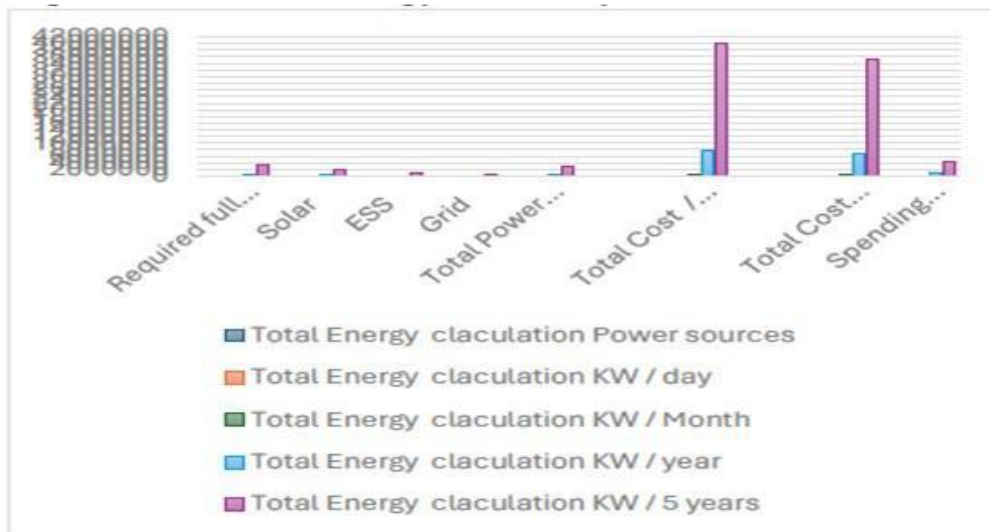


Fig: 18. Five-year energy contribution projections by source.

C. Comparative Performance Metrics

Table I compares key performance indicators between simulation predictions and hardware measurements. The close agreement across all four metrics confirms that the simulation model accurately represents physical system behavior and that control parameters transfer from simulation to hardware without significant re-tuning.

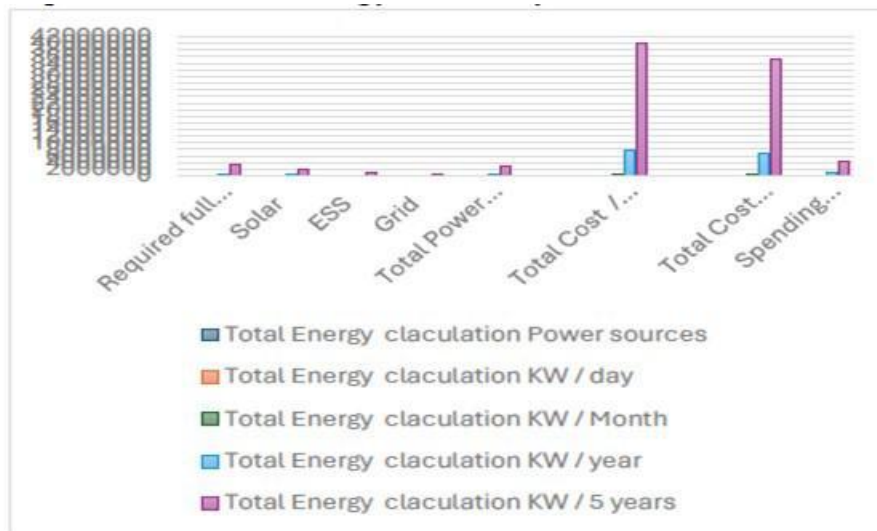


Fig: 19. Five-year cost and revenue projections.

TABLE I
PERFORMANCE METRICS COMPARISON (SIMULATION VS. HARDWARE)

Metric	Simulation	Hardware
Renewable Energy Utilization	88.6%	86.2%
ESS Round-Trip Efficiency	92.0%	89.5%
DC Bus Voltage Regulation	$\pm 2.1\%$	$\pm 3.5\%$
PCC Transition Time	< 150 ms	< 200 ms

TABLE I — Performance Metrics: Simulation vs. Hardware

The 2.4% difference in renewable utilization between simulation and hardware is attributable to real-world converter losses and sensor measurement tolerances not fully captured in the ideal switching model. The wider voltage regulation band in hardware reflects parasitic impedances in the physical DC bus wiring. Both deviations are within acceptable margins for a laboratory-scale prototype and do not affect the validity of the control architecture.

D. Economic Viability Analysis

The financial model, scaled to full-system deployment parameters, confirms the economic viability of the proposed architecture. Daily renewable generation covers the dominant share of station energy demand, with grid consumption limited to the backup contribution measured during hardware testing. Revenue from DC fast charging and battery swapping services combined substantially exceeds daily grid energy expenditure, establishing a positive net operating position from day one of deployment. Projected over five years, cumulative station profit confirms a compelling business case for deploying the proposed architecture at scale, with the economic outcome driven primarily by renewable utilization efficiency rather than tariff conditions — making the result robust across different grid pricing environments.

VII. CONCLUSION

This paper described the design, hardware implementation, and experimental validation of an IoT-enabled hybrid energy management system for EV charging stations. The central result is a working laboratory prototype — not a simulation study with hardware described in a future-work section — that runs all five IoT layers simultaneously and demonstrates automated solar PV, ESS, and grid dispatch through a PCC whose relay configuration changes in real time based on measured threshold crossings. Simulation predicted 88.6% renewable energy contribution; the prototype measured 86.2%. The 2.4 percentage point difference comes from converter conduction losses and sensor tolerances that the ideal switching model does not capture. That gap was anticipated; it does not change the conclusion that the control logic, threshold settings, and energy sizing transfer from simulation to hardware without re-tuning.

On the prototype, ESS SOC held within the 20–80% band across all test cycles — no deep discharge, no overcharge. DC bus voltage stayed within $\pm 3.5\%$ of 430V nominal through load steps and source switching events including the 50 kW DCFC initiation surge. Mode transitions completed in under 200 ms at every threshold crossing, with no supply gap recorded at the BSS or DCFC ports. The prototype handled both load types concurrently under the same dispatch framework. Published hardware implementations of hybrid EV charging architectures have addressed BSS or DCFC loads in isolation; this prototype managed both simultaneously. The system requires no utility-side infrastructure changes, which was a binding practical constraint in the design. What comes next is harder: V2G integration requires bidirectional grid certification that laboratory prototypes do not need, tariff-responsive scheduling requires real utility pricing data, and multi-station coordination introduces communication dependencies that single-station validation cannot expose. Those are the problems the present architecture is positioned to address.

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